Robotic Exoskeleton for Load Transportation

Software-centric intent recognition & sensor fusion for assistive control

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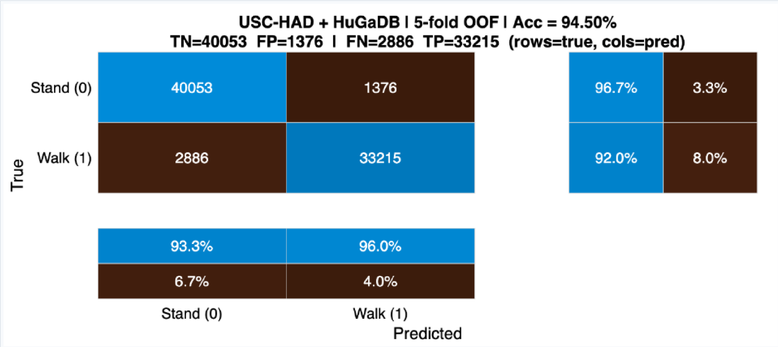
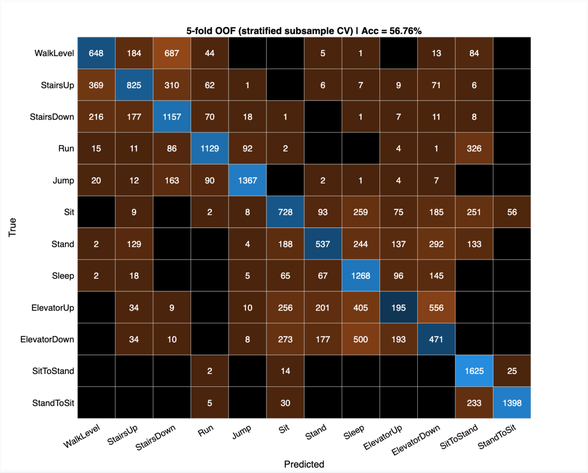
Binary SVM (merged, 5-fold OOF)
94.50%
USC-HAD + HuGaDB

12-class ECOC (5-fold OOF)
56.76%
Stratified subsample

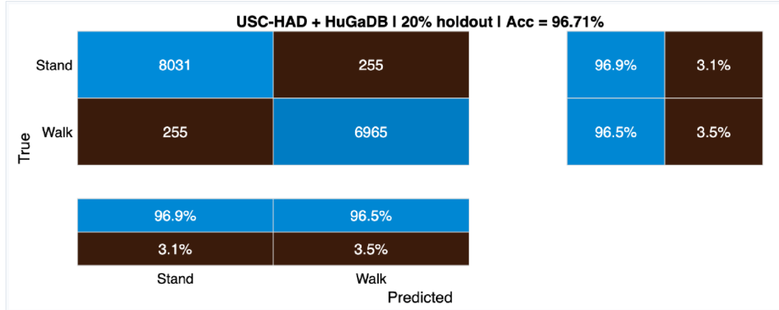
Binary LSTM (holdout)
96.71%
Not comparable to SVM OOF



Binary confusion matrix — merged SVM, five-fold OOF



Twelve-class ECOC — five-fold OOF



Binary LSTM — stratified 20% holdout

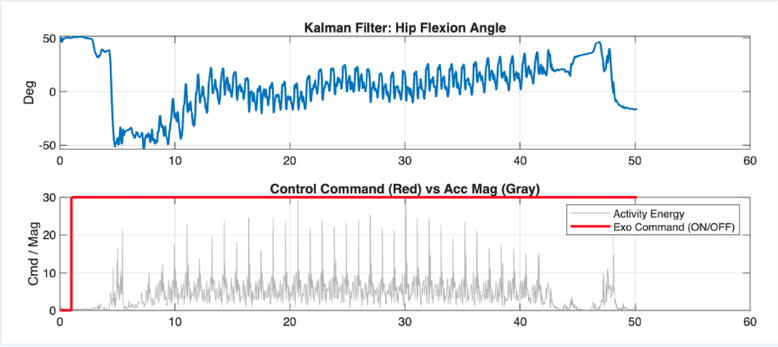
System overview

Sliding windows (1.0 s, 50% overlap, 100 Hz): **30-D** feature vector (five statistics per IMU slot × six slots). HuGaDB uses six streams; USC-HAD single-stream windows are zero-padded.

RBF SVM (`fitcsvm`) is the primary classifier; **multiclass ECOC** (12 labels) supports analysis; **FSM**: 3 consecutive walk-like predictions to assist, 5 to release—reducing chatter in simulation.

Real-time pipeline (simulation)

Kalman-based hip angle, classifier output, and FSM-latched assist command—stable engagement vs. rapid toggling.



Pipeline output: hip flexion, assist vs. acceleration magnitude, annotations when available

Binary SVM accuracy (ablation)

Training data	OOF acc.
USC-HAD only	**98.91%**
HuGaDB only	**88.60%**
USC-HAD + HuGaDB (merged)	**94.50%**

The $> 90\%$ target is met for USC-only and merged training; HuGaDB-only is harder (variability / imbalance). The **merged** model is the default for broader sensing layouts.

Multiclass & LSTM baselines

12-class ECOC: exploratory unified labels (harder than binary assist gating).

LSTM: sequence baseline on the same binary task; holdout metric is not directly comparable to SVM OOF.

Note: LSTM training was interrupted before the final epoch; see report.

Motivation

Heavy manual handling (> 20 kg) drives MSDs, back pain, and lost productivity. Passive or fixed-profile exoskeletons often mismatch user intent and context.

This work targets a **lightweight, software-based** module: real-time **locomotion vs. non-locomotion** gating using IMU data and classical ML—without EMG—for eventual hip/knee assistance during load transport.

Goals

- Kalman-based fusion for joint kinematics from a **three-phone IMU** prototype (CSV replay).
- **Binary** locomotion intent (locomotion-like vs. non-locomotion) with SVM features on **USC-HAD** and **HuGaDB**.
- FSM smoothing for stable assist commands; optional **binary LSTM** sequence baseline.
- Metrics synced with `generated_metrics.tex` / MATLAB exports.

Takeaways

- Reproducible **binary assist gating** on public IMU data with strong merged accuracy.
- **Kalman + SVM + FSM** integrated end-to-end in MATLAB for CSV-driven simulation.
- Next: hardware-in-the-loop, trials under load, aligned LSTM vs. SVM validation, end-to-end latency studies.

Code & report

https://github.com/Rex-Yim/AutomationForExoskeleton

Scalars and figures match the FYP report when `ExportMetricsForReport.m` has been run.